American International University-Bangladesh (AIUB)
Department of Computer Science
Faculty of Science &Technology (FST)
SPRING 25-26

Section: C
Software Quality Assurance and Testing

Smart Public Transport Ticketing and Fare Management System

A Report submitted

By

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Software Test Plan

for

< Smart Public Transport Ticketing and Fare Management System >

Version 1.0 approved

Prepared by <author>

<organization>

<date created>

Table of Contents

Revision History

Revision	Date	Updated by	Update Comments
0.1	2026.04.25	Asifur Rahman	Initial draft of test plan
0.2	2026.04.27	MD. Thamedul Islam shadin	Added system features and requirements
0.3	2026.04.29	Niladri Biswas Riva	Updated test cases and NFR section
0.4	2026.04.30	Sanjukta Das	Added risk management and references
0.5	2026.05.01	Shahriyer Islam	Final review and formatting

1. TEST PLAN IDENTIFIER:

SQT-SPTMS-01
(Software Quality Testing – Smart Public Transport Ticketing and Fare Management System – Version 1.0)

2. REFERENCES

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- bKash Developer Portal. *bKash Payment Gateway Integration Guide*. https://developer.bka.sh/
- Nagad API Documentation. *Digital Payment System Integration*. https://developer.nagad.com.bd/

3. INTRODUCTION

Background to the Problem

In modern urban environments, public transportation plays a crucial role in daily life. With the increasing number of passengers, traditional ticketing systems—such as paper tickets and manual fare collection—are becoming inefficient, time-consuming, and error-prone. These conventional methods often lead to long queues, fare miscalculations, ticket fraud, and poor passenger experience.

To overcome these limitations, many cities have started adopting digital ticketing systems. However, even existing smart ticketing solutions face several challenges, especially under high user demand. Systems often experience slow performance, incorrect fare calculations, seat allocation errors, and failures in real-time updates. These problems become more critical during peak hours when thousands of users access the system simultaneously.

Additionally, the integration of third-party services such as payment gateways, GPS tracking, and QR code validation increases system complexity. As a result, any failure in these components can disrupt the entire system. According to the project document, users are already facing issues like booking delays, payment failures, app crashes, and inaccurate tracking .

These issues reduce user trust, decrease system reliability, and negatively impact the overall efficiency of public transport management.

Solution to the Problem

To address these challenges, this project proposes a Smart Public Transport Ticketing and Fare Management System. The system is designed as a centralized, web-based and mobile-supported platform that automates ticket booking, fare calculation, and passenger management.

The proposed system will allow users to:

- Book tickets online
- Make secure digital payments
- View real-time seat availability
- Track transport vehicles using GPS
- Use QR codes for fast validation

From the management perspective, the system will:

- Ensure accurate fare calculation
- Maintain a centralized database of transactions
- Reduce human errors
- Improve transparency and reporting

This solution is appropriate because it enhances efficiency, reduces operational costs, and improves user satisfaction. It is also technically feasible, as it utilizes widely available technologies such as web applications, mobile platforms, and secure payment systems.

Existing systems mainly focus on either ticket booking or payment processing, but this proposed system integrates all functionalities—ticketing, fare management, tracking, and reporting—into a single platform. Therefore, it provides a more complete and scalable solution for modern public transportation systems.

Target Users & System Justification

The following stakeholders are the primary target users of the system and reasons why they should use our system:

1. Passengers (General Users)

Why they are targeted: Passengers are the primary end-users who face the most pain from the current manual system — wasted time, no seat guarantee, no bus location data.

Why use OUR system: Our platform provides a seamless, one-stop experience to search buses, book seats, pay digitally, and track the bus in real-time from their smartphone. Unlike other apps that may only offer one feature, ours integrates all three — booking, payment, and tracking — in a single platform tailored for Bangladesh's local payment methods (bKash, Nagad).

2. Bus Operators / Drivers

Why they are targeted: Drivers and operators need a fast and reliable way to verify passenger tickets and update trip statuses without manual checks.

Why use our system: The operator app provides a dedicated QR scanner for instant ticket verification, reducing boarding time and fraud. Trip status updates are automated and shared with passengers in real-time.

3. Transport Company / Admin

Why they are targeted: Transport companies need centralized control over routes, fleets, pricing, and revenue — something impossible with the current paper-based system.

Why use OUR system: The admin dashboard offers full control over routes, seats, and pricing, with automated reports on revenue and bookings. This eliminates revenue leakage and enables data-driven management.

4. Payment Service Providers (bKash, Nagad, etc.)

Why they are targeted: Digital payments are the backbone of the ticketing system, and integration with Bangladesh's most popular mobile money platforms is essential for adoption.

Why use OUR system: Our system integrates directly with local payment APIs, ensuring fast, secure, and verified transactions without the need for international card infrastructure.

5. GPS Service Providers

Why they are targeted: Real-time tracking is a key differentiator of our system. GPS providers supply the live data that powers the tracking feature.

Why use OUR system: The system is designed with a modular GPS integration layer, making it easy for GPS providers to plug in their data stream and serve millions of passengers with live bus locations.

4. REQUEIREMNT SPECIFICATION

4.1 System Features

ID	As a/An	I want to	So that	Acceptance Criteria (AC)
T1	User	Register using email or phone	I can create an account	Valid email/phone required; OTP verification; data stored successfully
T2	User	Login securely	I can access my account	Correct credentials login; invalid shows error; session created
T3	User	Logout	I can exit my account	Session terminated; redirected to login
T4	User	Verify OTP	My account stays secure	OTP sent; correct OTP accepted; expired rejected
T5	User	Reset password	I can recover my account	Reset link/OTP sent; new password validated; updated
T6	User	Manage profile	My information is updated	User edits data; validation applied; saved correctly
T7	User	Login from multiple devices	I can access anywhere	Multiple sessions allowed; sync maintained
T8	Admin	Assign roles	Users get proper access	Role assigned; access restricted; unauthorized blocked
T9	System	Verify account	Only valid users can use system	Email/OTP verification; unverified blocked
T10	User	Search buses	I can find available buses	Input route/time; results displayed correctly
T11	User	View seat availability	I can choose seats	Real-time seats shown; booked seats locked
T12	User	Select seats	I can reserve seat	Seat selection works; unavailable seats blocked
T13	User	Book ticket	I can confirm travel	Ticket generated; booking stored; confirmation shown
T14	User	Cancel ticket	I can change plans	Cancel option works; seat released; confirmation shown
T15	User	Reschedule ticket	I can adjust time	New schedule selected; booking updated
T16	System	Calculate fare	User sees correct price	Fare based on route/distance; correct total shown

T17	User	View booking history	I can track bookings	Past bookings visible; details shown
T18	User	Choose payment method	I can pay easily	bKash/Nagad/Card options; selection works
T19	System	Process payment	Transaction completes	Payment success handled; failure handled
T20	User	Retry payment	I can complete failed payment	Retry option shown; payment reprocessed
T21	System	Process refund	User gets money back	Refund initiated; correct amount returned
T22	User	View transactions	I can track payments	Transaction list shown; details accurate
T23	System	Generate receipt	User gets proof	Receipt generated; downloadable
T24	User	Track bus live	I know bus location	GPS shows real-time location; updates continuously
T25	User	See ETA	I know arrival time	ETA displayed; updates dynamically
T26	System	Notify delay	User stays informed	Delay alert sent; updated ETA shown
T27	User	View route map	I understand route	Map shows correct path
T28	System	Adjust route	Delay reduced	Route updated based on traffic
T29	System	Generate QR code	Ticket can be validated	Unique QR generated per ticket
T30	System	Scan QR	Ticket is verified	QR scan successful; valid ticket accepted
T31	System	Verify QR offline	Works without internet	QR validation works offline
T32	System	Prevent duplicate ticket	Avoid fraud	Same QR cannot be reused
T33	System	Send booking notification	User is informed	Booking confirmation sent
T34	System	Send payment alert	User knows status	Payment success/failure alert
T35	System	Send arrival alert	User is prepared	Arrival notification sent
T36	System	Send system alerts	User stays updated	Maintenance notification sent
T37	Admin	Manage schedules	System runs properly	Add/update/delete schedule works
T38	Admin	Manage seats	Capacity controlled	Seat count updated correctly
T39	Admin	Manage users	Control users	View/edit/delete users works

T40	Admin	View analytics	Monitor system	Reports generated; logs tracked

1. System Login and Authentication

Functional Requirements

1.1 The system shall allow users to log in using valid email/phone and password.
1.2 The system shall authenticate user credentials against the database.
1.3 The system shall display an error message for invalid login attempts.
1.4 The system shall support OTP-based authentication when required.
1.5 The system shall temporarily lock the account after multiple failed login attempts.
1.6 The system shall allow users to securely log out of the system.

Priority Level: High
Precondition: User must have a registered account.

2. User Registration

Functional Requirements

2.1 The system shall allow users to register using email or phone number.
2.2 The system shall validate all user inputs before registration.
2.3 The system shall send OTP for verification during registration.
2.4 The system shall store user information securely in the database.

Priority Level: High
Precondition: User must provide valid registration details.

3. User Profile Management

Functional Requirements

3.1 The system shall allow users to view their profile information.
3.2 The system shall allow users to update personal information.
3.3 The system shall validate updated data before saving.
3.4 The system shall allow users to reset or change passwords.

Priority Level: High
Precondition: User must be logged in.

4. Multi-Device Login Support

Functional Requirements

4.1 The system shall allow users to log in from multiple devices.
4.2 The system shall maintain session synchronization across devices.
4.3 The system shall allow users to log out from all devices.

Priority Level: Medium
Precondition: User must be authenticated.

5. Role-Based Access Control

Functional Requirements

5.1 The system shall define roles such as Admin and User.
5.2 The system shall restrict access based on roles.
5.3 The system shall prevent unauthorized actions.

Priority Level: High
Precondition: User must be logged in.

6. Bus Search System

Functional Requirements

6.1 The system shall allow users to search buses by route, time, and location.
6.2 The system shall display available buses based on search criteria.
6.3 The system shall support filtering of results.

Priority Level: High
Precondition: User must be logged in.

7. Seat Availability

Functional Requirements

7.1 The system shall display real-time seat availability.
7.2 The system shall update seat status dynamically.
7.3 The system shall prevent the selection of booked seats.

Priority Level: High
Precondition: Bus must be selected.

8. Seat Selection

Functional Requirements

8.1 The system shall allow users to select seats visually.
8.2 The system shall highlight selected seats.
8.3 The system shall confirm seat selection before booking.

Priority Level: High
Precondition: Seat availability must be displayed.

9. Ticket Booking

Functional Requirements

9.1 The system shall allow users to book tickets instantly.
9.2 The system shall generate a unique ticket ID.
9.3 The system shall store booking details in the database.
9.4 The system shall confirm booking after successful payment.

Priority Level: High
Precondition: User must select seat and proceed to booking.

10. Ticket Cancellation

Functional Requirements

10.1 The system shall allow users to cancel tickets.
10.2 The system shall update seat availability after cancellation.
10.3 The system shall initiate a refund process after cancellation.

Priority Level: High
Precondition: Ticket must be booked.

11. Ticket Rescheduling

Functional Requirements

11.1 The system shall allow users to reschedule tickets.
11.2 The system shall update new travel details.
11.3 The system shall replace old booking with new booking.

Priority Level: Medium
Precondition: Ticket must exist.

12. Fare Calculation

Functional Requirements

12.1 The system shall calculate fare based on route and distance.
12.2 The system shall display total fare before payment.
12.3 The system shall update the fare dynamically.

Priority Level: High
Precondition: Route must be selected.

13. Booking History

Functional Requirements

13.1 The system shall store all booking records.
13.2 The system shall allow users to view booking history.
13.3 The system shall display booking details.

Priority Level: Medium
Precondition: User must be logged in.

14. Payment System

Functional Requirements

14.1 The system shall support multiple payment methods (bKash, Nagad, Card).
14.2 The system shall process payments securely.
14.3 The system shall handle payment failures.
14.4 The system shall allow retry options.
14.5 The system shall generate payment receipts.
14.6 The system shall process refunds.

Priority Level: High
Precondition: User must confirm booking.

15. Transaction History

Functional Requirements

15.1 The system shall store transaction records.
15.2 The system shall allow users to view transaction history.

Priority Level: Medium
Precondition: User must be logged in.

16. Live Bus Tracking

Functional Requirements

16.1 The system shall provide live bus tracking using GPS.
16.2 The system shall update location in real-time.

Priority Level: Medium
Precondition: GPS must be active.

17. ETA System

Functional Requirements

17.1 The system shall display estimated arrival time (ETA).
17.2 The system shall update ETA dynamically.

Priority Level: Medium
Precondition: Tracking must be enabled.

18. Delay Notification

Functional Requirements

18.1 The system shall notify users of delays.
18.2 The system shall update delay information.

Priority Level: Medium
Precondition: System must detect delay.

19. Route Visualization

Functional Requirements

19.1 The system shall display routes on a map.
19.2 The system shall show the correct travel path.

20. Traffic Adjustment

Functional Requirements

20.1 The system shall adjust routes based on traffic conditions.
20.2 The system shall minimize delays.

21. QR Code Generation

Functional Requirements

21.1 The system shall generate unique QR code for tickets.

22. QR Validation

Functional Requirements

22.1 The system shall scan QR code.
22.2 The system shall validate tickets.

23. Offline QR

Functional Requirements

23.1 The system shall validate QR without internet.

24. Duplicate Prevention

Functional Requirements

24.1 The system shall prevent duplicate ticket usage.

25. Booking Notification

Functional Requirements
25.1 The system shall send booking confirmation notifications to users.

26. Payment Alerts

Functional Requirements
26.1 The system shall notify users about payment success or failure.

27. Bus Arrival Alerts

Functional Requirements
27.1 The system shall send alerts before bus arrival.

28. System Notifications

Functional Requirements
28.1 The system shall notify users about maintenance or updates.

29. Bus Schedule Management

Functional Requirements
29.1 The system shall allow admin to add, update, and delete bus schedules.

30. Seat Management

Functional Requirements
30.1 The system shall allow admin to manage seat capacity.

31. User Management

Functional Requirements
31.1 The system shall allow admin to view, edit, and delete users.

32. Analytics & Reports

Functional Requirements
32.1 The system shall generate reports on revenue and usage.

33. System Monitoring

Functional Requirements
33.1 The system shall monitor system performance and activities.

34. Error Logging

Functional Requirements
34.1 The system shall record and store system errors.

35. Security Monitoring

Functional Requirements
35.1 The system shall detect and prevent unauthorized access.

36. Data Backup

Functional Requirements
36.1 The system shall perform regular data backup.

37. Data Recovery

Functional Requirements
37.1 The system shall restore data in case of failure.

38. Performance Optimization

Functional Requirements
38.1 The system shall optimize performance during high load.

39. System Availability

Functional Requirements
39.1 The system shall ensure continuous availability.

40. Audit Logs

Functional Requirements

40.1 The system shall maintain logs of all system activities.

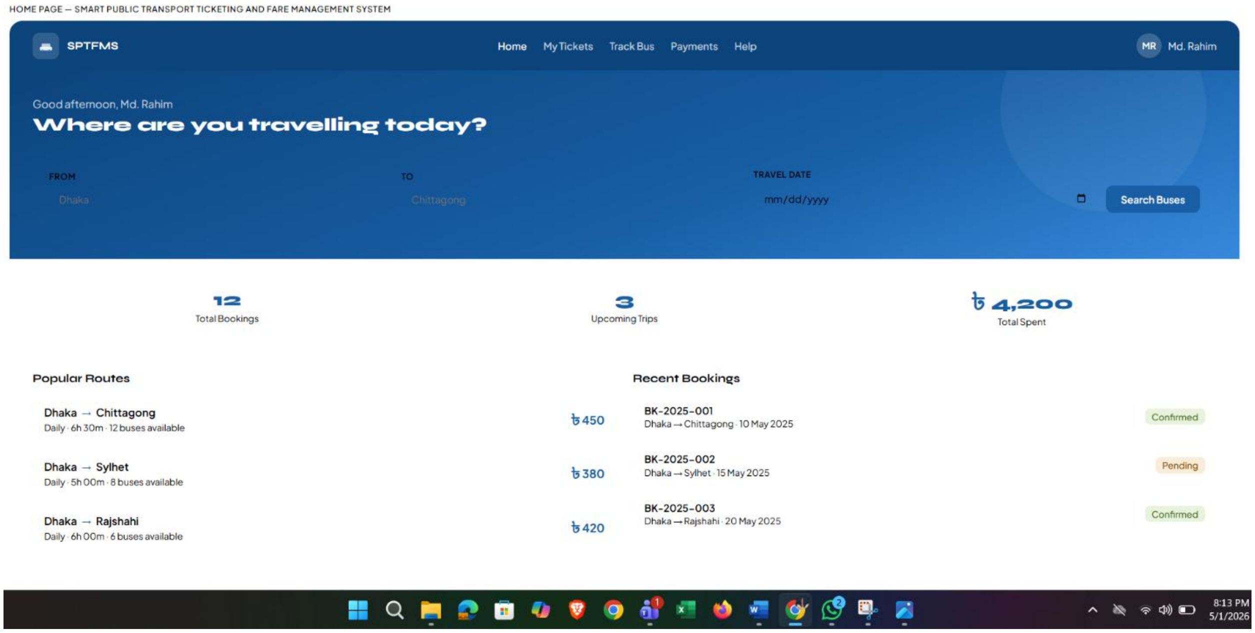
4.2 System Quality Attributes

NFR	Type	Actual Requirement
Q1	Performance	System response time must be less than 3 seconds for user actions
Q2	Performance	System must support thousands of concurrent users without failure
Q3	Reliability	System must be available 24/7 except scheduled maintenance
Q4	Reliability	System must ensure 99% uptime
Q5	Security	All user data must be encrypted (SSL/TLS)
Q6	Security	Secure login with password and OTP verification
Q7	Security	Role-based access control must be enforced
Q8	Data	System must ensure accurate and consistent data storage
Q9	Data	System must perform regular automatic backups
Q10	Data	System must recover data in case of system failure
Q11	Usability	Users should complete ticket booking within 3 minutes
Q12	Usability	System interface must be user-friendly and easy to navigate
Q13	Compatibility	System must work on web and mobile devices
Q14	Compatibility	System must support Chrome, Firefox, Edge
Q15	Maintainability	System should be easy to update and maintain
Q16	Maintainability	System components must be modular and reusable
Q17	Monitoring	System must record all activities and errors
Q18	Reliability	System must handle errors

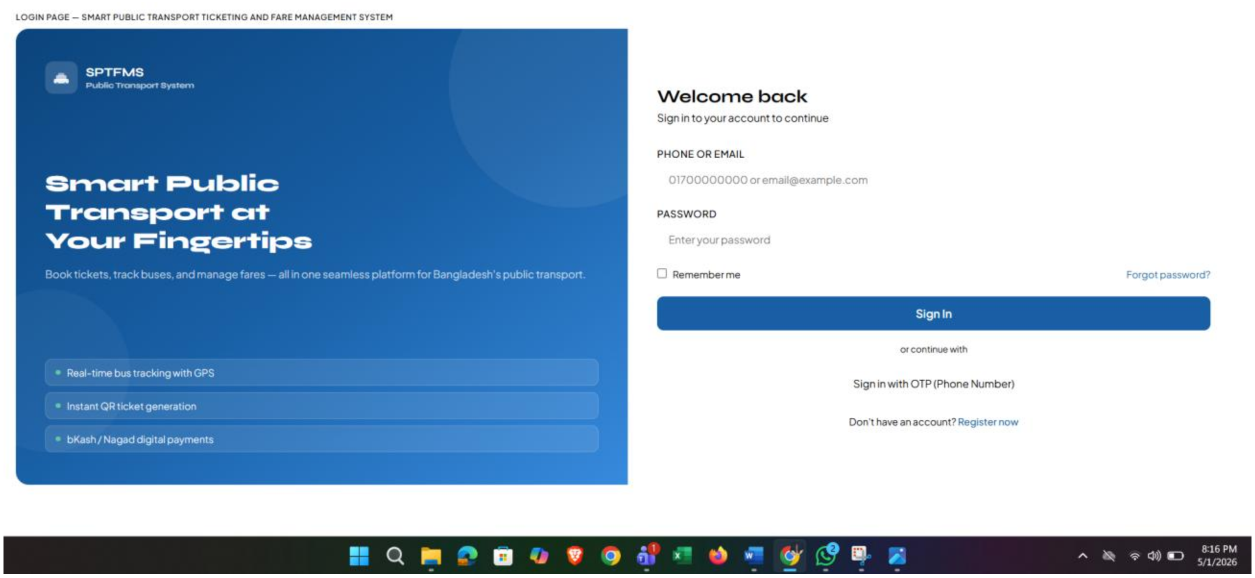
		without crashing
Q19	Performance	Notifications must be delivered instantly
Q20	Security	Payment transactions must be secure and verified
Q21	Performance	Bus tracking must update location in real-time
Q22	Performance	System must handle peak load efficiently
Q23	Reliability	System must recover quickly after failure
Q24	Security	System must maintain logs of all user/admin actions
Q25	Security	Sessions must expire after inactivity

4.3 System Interface

Home Page:



Login Page:



4.4 Project Requirements

Cost Category	Estimated Cost (BDT)	Details
Human Resources (Developers, Testers, PM)	3,60,000	5 members × 4 months × 18,000 avg.

Cost Category	Estimated Cost (BDT)	Details
Infrastructure & Server (Cloud Hosting)	60,000	AWS/Azure (scalable hosting + backup)
Third-Party APIs (Payment, GPS, QR)	50,000	bKash/Nagad, Google Maps, QR libs
Testing Tools & Licenses	40,000	Selenium, JMeter, Postman Pro
Development Tools & Software	35,000	IDEs, DB tools, design software
Documentation & Miscellaneous	20,000	Printing, meetings, contingency
TOTAL	**5,65,000 BDT**	**Approved budget ceiling**

Time Estimation

Phase	Duration	Key Activities
Requirements & Planning	2 weeks	Stakeholder interviews, SRS, use-case modeling
System Design	2 weeks	Architecture, DB schema, UI wireframes
Development – Module 1 (Auth, Booking)	3 weeks	Login, registration, seat selection, ticketing
Development – Module 2 (Payment, QR, GPS)	3 weeks	bKash/Nagad, QR generation, live tracking
Development – Module 3 (Admin, Reports)	2 weeks	Admin dashboard, analytics, notifications
Testing (Unit, Integration, Acceptance)	2 weeks	Test design, execution, defect fixing

Phase	Duration	Key Activities
Deployment & Training	1 week	Production deploy, user training, handover
TOTAL	**3.5 Months(Almost)**	**Estimated completion**

5. FEATURES NOT TO BE TESTED

The following areas are not specifically or directly addressed in the test plan for the Smart Public Transport Ticketing and Fare Management System. Any testing coverage in these areas is incidental, arising as a by-product of other testing activities.

Third-Party Payment Gateway Internal Processing

- bKash, Nagad, and card payment platforms have their own internal processing logic, fraud detection, and compliance mechanisms. These are completely under the control and responsibility of the respective payment providers and are outside the scope of this project. The system will only verify whether the payment API returns a success or failure response; the internal transaction processing of those platforms will not be tested.

Mobile Operating System and Device Hardware

- The behavior of the application across different Android or iOS device hardware (screen resolution, RAM, GPU) and OS-level features (notifications daemon, background processes) is controlled by the end user's device and OS vendor. Testing of OS-specific rendering or hardware behavior is the responsibility of the device manufacturer and OS vendor.

Third-Party GPS and Mapping Services

- Google Maps API and GPS satellite-level accuracy are outside the scope of this project. The system will test whether GPS data is received and displayed correctly on the interface, but accuracy, latency, or outages of the GPS/mapping service itself will not be tested.

Network Infrastructure and ISP Connectivity

- The reliability or speed of the end user's internet connection, mobile network provider, or ISP is not within the control of the project team. Testing will assume a stable connection is available. Network failure scenarios will only be tested to the extent of verifying the system's error-handling responses.

Third-Party QR Code Library Internal Encoding

- The internal encoding and decoding algorithm of the QR code generation library used is a third-party component. Testing will verify that the QR code produced functions correctly for ticket validation, but the internal library implementation will not be tested.

Customer-Side Data Extraction and Reporting Tools

- If end users or administrators choose to export data (e.g., transaction logs, booking records) and process them in external spreadsheet applications (e.g., Microsoft Excel, Google Sheets), those applications are entirely under the control of the customer. The system will provide the necessary data export format (CSV/JSON). Testing of any customer-side tools or analyses is the responsibility of the application user/administrator.

Email and SMS Delivery Infrastructure

- The delivery of reliability of OTP messages and booking confirmation notifications through third-party SMS gateways (e.g., SSL Wireless) and email providers (e.g., SendGrid) is not within the scope of testing. The system will be tested to confirm that the notification trigger is sent correctly; actual delivery at the recipient device is the responsibility of the third-party gateway.

Admin Hardware and Operating Environment

- The hardware, OS, and browser environment used by system administrators to access the admin dashboard are managed by the client organization's IT department. The project will provide supported browser compatibility (Chrome, Firefox, Edge) but testing proprietary or unsupported admin environments is the client's responsibility.

6. TESTING APPROACH

6.1 Testing Levels

The testing for the Smart Public Transport Ticketing and Fare Management System consists of three primary test levels: Unit Testing, System/Integration Testing (combined), and Acceptance Testing. Given the budget and staffing constraints of the project, the test manager will lead most testing activities with participation from the development team.

Unit Testing

- Unit Testing will be conducted by the individual developer responsible for each module (e.g., authentication, fare calculation, QR generation, payment processing).
- Each developer must produce proof of unit testing — including a test case list, sample output logs, data printouts, and any defect information — before the unit is accepted by the team leader.
- Unit tests must be approved by the development team leader prior to progressing to the next testing level.
- All unit test documentation will be handed over to the QA tester for reference during integration testing.
- No unit will be passed to System/Integration testing if it has any Critical defect outstanding.

System / Integration Testing (Combined)

- System/Integration Testing will be performed by the Test Manager and Development Team Leader, with assistance from individual developers as required.
- A module will enter System/Integration testing only after all Critical defects from unit testing have been resolved.
- A module may carry up to two Major defects into integration testing, provided those defects do not impede testing (i.e., a workaround exists, and testing of other features is not blocked).
- Integration testing will focus on verifying the interaction between modules: authentication ↔ booking, booking ↔ payment, payment ↔ QR generation, GPS ↔ tracking display, and admin ↔ reporting.

Acceptance Testing

- Acceptance Testing will be performed by actual end users (passengers and transport administrators) with the assistance of the Test Manager and Development Team Leader.
- Acceptance testing will run in parallel with the existing manual ticketing/fare collection process for a period of one month after the completion of System/Integration Testing. This parallel run ensures that the new system produces results consistent with the existing process before full cutover.
- Acceptance test scenarios will be derived directly from the User Stories (T1–T40) defined in the Requirement Specification section.
- Acceptance is granted only when all high-priority user stories pass their defined Acceptance Criteria (AC) without Critical or Major defects.

6.2 Test Tools

The following automated and manual testing tools are used in this project:

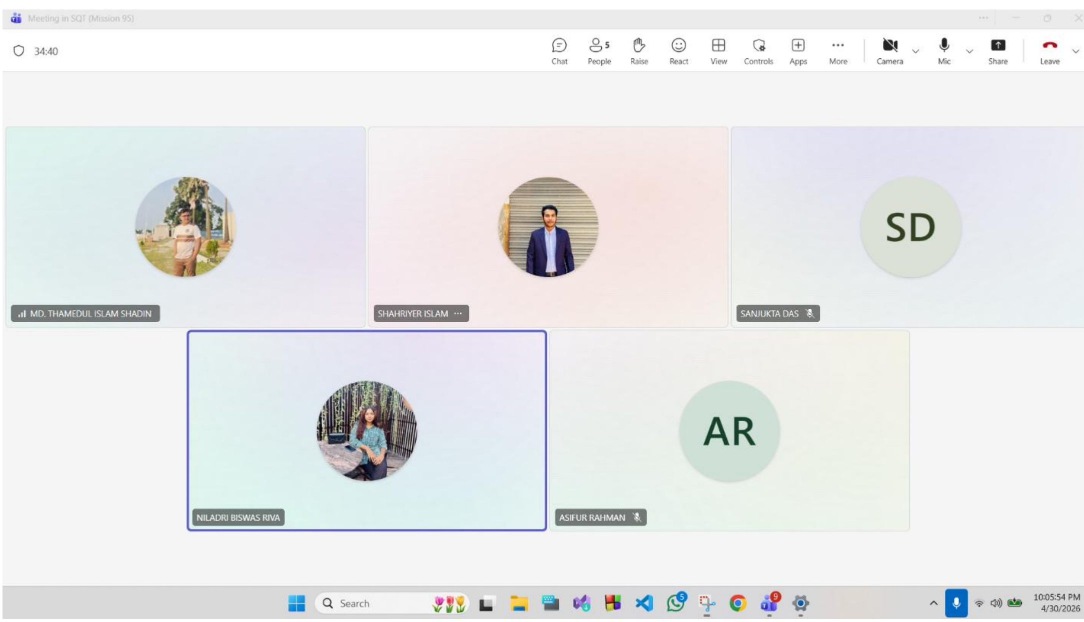
- Selenium is used for automating web-based functional testing such as login, ticket booking, and payment workflows.
- Postman (Manual + Automated) is used for API testing and automation of backend services including authentication, payment processing, QR generation, and GPS tracking.
- Apache JMeter is used for performance and load testing to evaluate system behavior under high user traffic conditions.
- Appium is used for mobile application testing to automate user interactions such as OTP login, QR code validation, and notifications.
- TestNG is used as a testing framework to manage test execution and generate test reports.

6.3 Meetings

The test team meets once a every week to evaluate progress to date and to identify error trends and problems as early as possible. The test team leader meets with development and the project manager once every weeks as well. These meetings scheduled are held on different weeks. Additional meetings called as required for emergency situations in Microsoft Teams.

Meeting Date	Signature				
25.3.2026	Asif	Niladri	Sanjukta	Shadin	Shariyer
1.4.2026	Asif	Niladri	Sanjukta	Shadin	Shariyer
8.4.2026	Asif	Niladri	Sanjukta	Shadin	Shariyer
15.4.2026	Asif	Niladri	Sanjukta	Shadin	Shariyer
22-4-2026	Asif	Niladri	Sanjukta	Shadin	Shariyer

Additional meeting for Emergency case:



7. TEST CASES/TEST ITEMS

1. User Registration

Project Name: **Smart Public Transport Ticketing and Fare Management System**	Test Designed by: Niladri Biswas
Test Case ID: FR_1	Test Designed date:01-05-2026

Test Priority (Low, Medium, High): High			Test Executed by: Niladri Biswas	
Module Name: Registration Session			Test Execution date: 05-05-2026	
Test Title: Registered with valid information				
Description: Test Bus Ticket Registration page validation				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: niladri biswas Password: 231	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

2. User Login

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Niladri Biswas	
Test Case ID: FR_2			Test Designed date: 01-05-2026	
Test Priority (Low, Medium, High): High			Test Executed by: Niladri Biswas	
Module Name: Login Session			Test Execution date:05-05-2026	
Test Title: verify login with valid username and password				
Description: Test website login page				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: niladri biswas Password: 321	User should login into the application	As expected,	Pass

Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.

3. Bus Search System

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Niladri Biswas	
Test Case ID: FR_10			Test Designed date:02-05-2026	
Test Priority (Low, Medium, High): High			Test Executed by: Niladri Biswas	
Module Name: Bus Searching			Test Execution date: 06-05-2026	
Test Title: Bus searching to website page				
Description: Test website page to searching bus				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status:
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: niladri biswas Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

4. Ticket Booking

Project Name: **Smart Public Transport Ticketing and Fare Management System**	Test Designed by: Niladri Biswas
Test Case ID: FR_13	Test Designed date: 02-05-2026
Test Priority (Low, Medium, High): High	Test Executed by: Niladri Biswas
Module Name: Ticket booking Session	Test Execution date:06-05-2026
Test Title: Ticket booking in website page	
Description: Test website page to booking ticket	
Precondition (If any): User must have valid username and password	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: niladri biswas Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

5. Ticket Rescheduling

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Sanjukta Das	
Test Case ID: FR_15			Test Designed date:02-05-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Sanjukta Das	
Module Name: Rescheduling ticket on website			Test Execution date:06-05-2026	
Test Title: Test website page to rescheduling				
Description: Test website page to rescheduling ticket				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: sanjukta das Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

6. Payment System

Project Name: **Smart Public Transport Ticketing and Fare Management System**	Test Designed by: Sanjukta Das
Test Case ID: FR_18	Test Designed date: 02-05-2026

Test Priority (Low, Medium, High): High			Test Executed by: Sanjukta Das	
Module Name: Payment System Session			Test Execution date:06-05-2026	
Test Title: verify payment system				
Description: Test website page to payment system				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: sanjukta das Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

7. Live Bus Tracking

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Sanjukta Das	
Test Case ID: FR_24			Test Designed date: 03-06-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Sanjukta Das	
Module Name: Live Bus Tracking session			Test Execution date: 06-05-2026	
Test Title: verify live bus tracking				
Description: Test website page to live bus tracking				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: 99999999999 Password: 321	User should login into the application	As expected,	Pass

Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.

8. Delay Notification

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Sanjukta Das	
Test Case ID: FR_26			Test Designed date:03-06-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Sanjukta Das	
Module Name: Delay Notification Session			Test Execution date: 06-05-2026	
Test Title: verify notification				
Description: Test website page to notification				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: sanjukta das Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

9. Traffic Adjustment

Project Name: **Smart Public Transport Ticketing and Fare Management System**	Test Designed by: Md. Thamedul Islam
Test Case ID: FR_28	Test Designed date: 04-05-2026
Test Priority (Low, Medium, High): Medium	Test Executed by: Md. Thamedul Islam
Module Name: Traffic Adjustment Session	Test Execution date:06-05-2026
Test Title: Verify Traffic adjustment	

Description: Test website page to traffic adjustment	
Precondition (If any): User must have valid username and password	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: Shadin Password: 321	User should login into the application	Not expected,	Fail
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

10.**QR Code Generation**

Project Name: **Smart Public Transport Ticketing and Fare Management System**	Test Designed by: Md. Thamedul Islam
Test Case ID: FR_29	Test Designed date: 04-05-2026
Test Priority (Low, Medium, High): Medium	Test Executed by: Md. Thamedul Islam
Module Name: QR code generate Session	Test Execution date: 06-05-2026
Test Title: verify QR code generate with valid username and password	
Description: Test website page to QR code generate	
Precondition (If any): User must have valid username and password	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: Shadin Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

11.QR Validation

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Md. Thamedul Islam	
Test Case ID: FR_30			Test Designed date:02-05-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Md. Thamedul Islam	
Module Name: QR validation			Test Execution date: 06-05-2026	
Test Title: verify QR validation with valid username and password				
Description: Test website page to QR validation				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: Shadin Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

12. Offline QR

Project Name: **Smart Public Transport Ticketing and Fare Management System**	Test Designed by: Md. Thamedul Islam
Test Case ID: FR_31	Test Designed date:02-05-2026
Test Priority (Low, Medium, High): Medium	Test Executed by: Md. Thamedul Islam
Module Name: Offline QR Session	Test Execution date: 06-05-2026
Test Title: verify offline QR with valid username and password	
Description: Test website page to offline QR	

Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: shadin Password: 321	User should login into the application	Not expected,	Fail
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

13. Booking Notification

Project Name: **Smart Public Transport Ticketing and Fare Management System**	Test Designed by: Asifur Rahman
Test Case ID: FR_33	Test Designed date: 03-05-2026
Test Priority (Low, Medium, High): Medium	Test Executed by: Asifur Rahman
Module Name: Booking Notification Session	Test Execution date: 06-05-2026
Test Title: verify booking notification with valid username and password	
Description: Test website page to booking notification	

Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: asifur Password: 321	User should login into the application	Not expected,	Fail
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

14. Payment Alerts

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Asifur Rahman	
Test Case ID: FR_34			Test Designed date: 03-05-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Asifur Rahman	
Module Name: Payment alerts Session			Test Execution date: 06-05-2026	
Test Title: verify payment alerts with valid username and password				
Description: Test website page to payment alerts				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: asifur Password: 321	User should login into the application	Not expected,	Fail
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

15. System Monitoring

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Asifur Rahman	
Test Case ID: FR_36			Test Designed date: 03-05-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Asifur Rahman	
Module Name: System monitoring Session			Test Execution date: 06-05-2026	
Test Title: Monitoring system				
Description: Test system monitoring				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: asifur Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

16.Error Logging

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Asifur Rahman	
Test Case ID: FR_1			Test Designed date: 03-05-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Asifur Rahman	
Module Name: Error Login Session			Test Execution date: 06-05-2026	
Test Title: Error login with valid username and password				
Description: Test website error login				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: asifur Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

17.Security Monitoring

Project Name: **Smart Public Transport Ticketing and Fare Management System**	Test Designed by: Shahriyer Islam
Test Case ID: FR_40	Test Designed date: 03-05-2026
Test Priority (Low, Medium, High): Medium	Test Executed by: Shahriyer Islam
Module Name: Security monitoring Session	Test Execution date:07-05-2025

Test Title: Security monitoring				
Description: Test website security monitoring				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: shaheiyer Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

18. Seat Management

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Shahriyer Islam	
Test Case ID: FR_38			Test Designed date: 03-05-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Shahriyer Islam	
Module Name: seat management Session			Test Execution date: 07-05-2025	
Test Title: seat management with valid username and password				
Description: Test website page to seat management				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: 99999999999 Password: 321	User should login into the application	Not expected,	Suspended
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

19. Multi-device logging support

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Shahriyer Islam	
Test Case ID: FR_7			Test Designed date: 03-05-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Shahriyer Islam	
Module Name: Multiple device Login Session			Test Execution date: 07-05-2025	
Test Title: verify multiple device login with valid username and password				
Description: Test website multiple device login page				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: shahriyer Password: 321	User should login into the application	Not expected,	Fail
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

20.**Log Out**

Project Name: **Smart Public Transport Ticketing and Fare Management System**			Test Designed by: Shahriyer Islam	
Test Case ID: FR_3			Test Designed date: 02-05-2026	
Test Priority (Low, Medium, High): Medium			Test Executed by: Shahriyer Islam	
Module Name: Logout Session			Test Execution date:05-05-2026	
Test Title: Logout Session				
Description: Test website logout page				
Precondition (If any): User is currently logged in				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Go to the website 2. Enter username 3. Enter password 4. Click submit	Username: shahriyer Password: 321	User should login into the application	As expected,	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

8. ITEM PASS/FAIL CRITERIA

TID	Module Name	Status
1	Registration Session	Pass
2	Login Session	Pass
10	Bus Searching	Pass
13	Ticket booking Session	Pass
15	Rescheduling ticket on website	Pass
18	Payment System Session	Pass
24	Live Bus Tracking session	Pass
26	Delay Notification Session	Pass
28	Traffic Adjustment Session	Fail
29	QR code generate Session	Pass
30	QR validation	Pass
31	Offline QR Session	Fail
33	Booking Notification Session	Fail
34	Payment alerts Session	Fail
36	System monitoring Session	Pass
1	Error Login Session	Pass
40	Security monitoring Session	Pass

38	seat management Session	Suspended
7	Multiple device Login Session	Fail
3	Logout Session	Pass

9. STAFFING AND TRAINING NEEDS

Name	Designation
Asifur Rahman	Junior Test Analyst
Md. Thamedul Islam Shadin	Test Engineer
Niladri Biswas Riva	QA Analyst
Sanjukta Das	Software Tester
Shahriyer Islam	System Tester

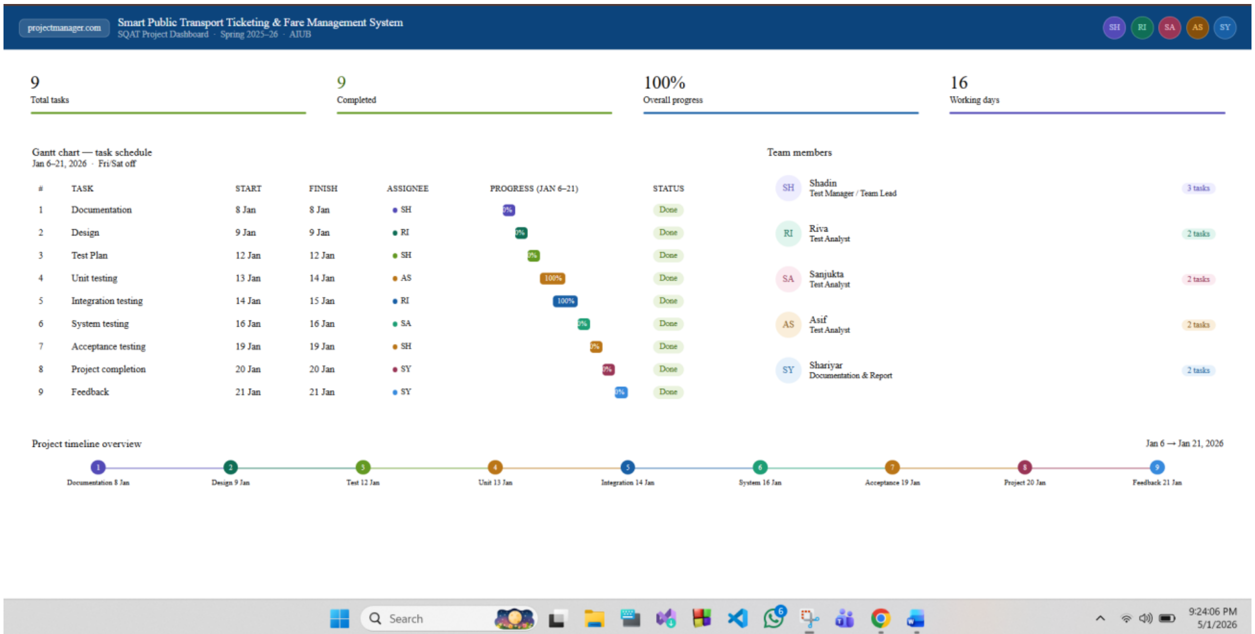
10. RESPONSIBILITIES

The following matrix defines the responsibilities of each team member across all testing and project activities.

Activity / Responsibility	Shadin (TM)	Riva (PM)	Sanjukta (Dev)	Asif (Test)	Shariyar (Doc)
TM = Test Manager PM = Project Manager Dev = Developer Test = Tester Doc = Documentation	TM	PM	Dev	Test	Doc
Acceptance test Documentation & Execution	✓	✓		✓	✓
System/Integration test Documentation & Exec.	✓		✓	✓	
Unit test documentation & execution	✓		✓	✓	
System Design Reviews	✓	✓	✓	✓	✓
Detail Design Reviews	✓	✓	✓	✓	
Test procedures and rules	✓	✓	✓	✓	
Screen & Report prototype reviews			✓	✓	✓
Change Control and regression testing	✓	✓	✓	✓	✓

11. TESTING SCHEDULE

Tool used: www.projectmanager.com



12. PLANNING RISKS AND CONTINGENCIES

Risk Description	Probability	Impact	Mitigation Plan
Payment Gateway Failure: Payment APIs (bKash/Nagad/Card) may fail or behave differently in a sandbox vs live environment.	Catastrophic	Critical	Use sandbox testing and mock responses. Perform small real transactions (e.g., 10 BDT) before deployment.
High System Load: System may crash or slow down during peak hours due to many users.	Catastrophic	Critical	Implement load balancing and optimize database queries. Use stress testing before deployment.
Security Vulnerabilities: Risk of data breach, SQL injection, or unauthorized access.	Moderate	Critical	Use OWASP security testing, SSL/TLS encryption, and secure authentication mechanisms.
GPS Tracking Inaccuracy: Live bus tracking may show incorrect location due to API issues.	Moderate	Catastrophic	Use reliable GPS APIs and implement fallback mechanisms for location updates.
Payment Failure	Moderate	Catastrophic	Implement retry

Handling Issues: Transactions may fail without proper handling or retry option.			mechanism and proper error handling for failed payments.
QR Code Validation Failure: QR code may not scan properly or may be duplicated.	Moderate	Catastrophic	Use unique encrypted QR codes and implement duplicate detection system.
Database Inconsistency: Seat booking or payment data may become inconsistent.	Moderate	Critical	Use transaction management and database locking to ensure data integrity.
Browser Compatibility Issues: System may not work properly on all browsers or devices.	Moderate	Minor	Test using Selenium across Chrome, Firefox, Edge and perform manual mobile testing.
Network Failure: Users may face issues due to poor internet connection.	Moderate	Moderate	Implement of retry mechanisms and offline support (QR validation).
Incomplete Requirements: Some system features may be unclear or missing.	Moderate	Catastrophic	Freeze requirements early and conduct regular review meetings with stakeholders.
Third-Party API Dependency: System depends on external APIs (payment, GPS).	Catastrophic	Catastrophic	Use backup APIs or fallback logic in case of failure.
Team Availability Issues: Team members may not be available due to workload.	Moderate	Moderate	Crosstrain team members and maintain proper documentation.

13. APROVALS

Role	Name
Project Sponsor	Asifur Rahman
Development Manager	Md. Thamedul Islam Shadin
Project Manager	Niladri Biswas Riva
Test Manager	Sanjukta Das
Development Team Manager	Shahriyer Islam
Quality Assurance Lead	Asifur Rahman
System Administrator	Md. Thamedul Islam Shadin